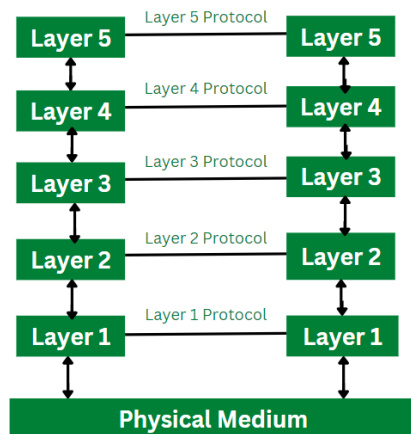


PART A

1. Illustrate layering principle with necessary diagrams.

Every network consists of a specific number of functions, layers, and tasks to perform. Layered Architecture in a computer network is defined as a model where a whole network process is divided into various smaller sub-tasks. These divided sub-tasks are then assigned to a specific layer to perform only the dedicated tasks. A single layer performs only a specific type of task. To run the application and provide all types of services to clients a lower layer adds its services to the higher layer present above it. Therefore layered architecture provides interactions between the sub-systems. If any type of modification is done in one layer it does not affect the next layer.



Features of Layered Architecture

Use of Layered architecture in computer network provides with the feature of modularity and distinct interfaces.

Layered architecture ensures independence between layers, by offering services to higher layers from the lower layers and without specifying how this services are implemented.

Layered architecture segments as larger and unmanageable design into small sub tasks.

In layer architecture every network has different number of functions, layers and content.

In layered architecture, the physical route provides with communication which is available under the layer 1.

In layered architecture, the implementation done by one layer can be modified by the another layer.

Need of Layered Architecture

Divide and Conquer Approach: Layered architecture supports divide and conquer approach. The unmanageable and complex task is further divided into smaller sub tasks. Each sub task is then carried out by the different layer. Therefore using this approach reduces the complexity of the problem or design process.

Easy to Modify: The layers are independent of each other in layered architecture. If any sudden change occurs in the implementation of one layer, it can be changed. This change does not affect the working of other layers involved in the task. Therefore layered architectures are required to perform any sudden update or change.

Modularity: Layered architecture is more modular as compared to other architecture models in computer network. Modularity provides with more independence between the layers and are more easier to understand.

Easy to Test: Each layer in layered architecture performs a different and dedicated task. Therefore each layer can be analyzed and tested individually. It helps to analyze the problem and solve them more efficiently as compared to solving all the problems at a time.

2. What is the propagation delay if distance between two points is 12,000 km?

Assume propagation speed to be 2.4×10^8 m/s in cable.

Distance = 12,000 km

Propagation speed = 2.4×10^8 m/s

Propagation delay = Distance / Propagation speed

= $12,000 \text{ km} / 2.4 \times 10^8 \text{ m/s}$

= 50ms

3. Show how the data field is managed in the Ethernet frame format.

Ethernet is a widely used networking technology that defines the framing of data for communication between devices on a local area network (LAN). The Ethernet frame format includes several fields, and the data field is one of them. Here's an overview of how the data field is managed in the Ethernet frame:

PREAMBLE	S F D	DESTINATION ADDRESS	SOURCE ADDRESS	LENGTH	DATA	CRC
7 Bytes	1 Byte	6 Bytes	6 Bytes	2 Bytes	46 - 1500 Bytes	4 Bytes

IEEE 802.3 ETHERNET Frame Format

Data – This is the place where actual data is inserted, also known as Payload. Both IP header and data will be inserted here if Internet Protocol is used over Ethernet. The maximum data present may be as long as 1500 Bytes. In case data length is less than minimum length i.e. 46 bytes, then padding 0's is added to meet the minimum possible length.

4. Differentiate between bridges and switches.

BRIDGE	SWITCH
Bridge works in data link layer of OSI model	Switch works in data link layer and network layer of <u>OSI model</u> .
The bridge has only two ports	Switch has multiple numbers of ports.
The bridge is used to connect two LAN segment using the same <u>topology</u> .	The switch is used to connect devices to the same network.
The bridge can operate only in <u>half-duplex</u> mode	Switch can operate in both <u>half duplex</u> and <u>full duplex mode</u> .
The performance of bridge is slower than switch	The performance of Switch is faster than a bridge.

5. Illustrate the optimality principle in routing.

The Optimality Principle

- It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.

- Eg: Let us call the part of the route from I to J as r1 and the rest of the route as r2.

. If a route better than r2 existed from J to K, it could be concatenated with r1 to improve the route from I to K, contradicting our statement that r1r2 is optimal.

6. Identify two methods by which multicast routing is handled by network layer.

Multicast routing in the network layer involves the efficient delivery of data from one source to multiple destinations. Two commonly used methods for handling multicast routing are:

Source-Based Tree (SBT):

In the Source-Based Tree approach, a separate distribution tree is built for each multicast source in the network. This means that for every source of multicast traffic, a tree is created to reach all the intended receivers. The tree spans the entire network from the source to the receivers.

Protocols that use the Source-Based Tree method include Protocol Independent Multicast (PIM) Sparse Mode (PIM-SM) and Bidirectional PIM (Bidir-PIM). PIM-SM, for example, builds a shared distribution tree rooted at the source, and routers prune branches of the tree where there are no active receivers.

One advantage of Source-Based Trees is that they can be optimized for the specific multicast group and source. However, this method may lead to redundant tree structures in the network, and it can be resource-intensive.

Shared Tree (ST):

In the Shared Tree approach, a single distribution tree is shared among multiple sources and their corresponding receivers. This tree is often referred to as the "rendezvous point (RP) tree" or "core-based tree." All multicast traffic is forwarded along this shared tree, and receivers join the tree at the closest point to them.

Protocol Independent Multicast Dense Mode (PIM-DM) and PIM-SM with shared trees are examples of protocols that use the Shared Tree approach. PIM-DM floods multicast traffic throughout the network, while PIM-SM uses an RP to create a shared tree.

Shared Trees can be more efficient in terms of resource utilization because multiple sources can share the same distribution tree. However, there may be suboptimal routing for some sources and receivers, as they all use the same shared path.

Networks often use a combination of these methods depending on factors such as the number of sources, the distribution of receivers, and the overall network topology. The choice between Source-Based Trees and Shared Trees depends on the specific requirements and characteristics of the multicast group and the network architecture.

7. In IP, the checksum covers only the header and not the data. Identify the reason for selecting this design.

In the Internet Protocol (IP), the checksum is used to provide a basic level of error detection for the IP header but not the data payload. The primary reason for this design choice is to minimize processing overhead and improve efficiency.

The IP header checksum is calculated over the fields of the IP header, including source and destination IP addresses, protocol type, Time-to-Live (TTL), and other header fields. The purpose of the checksum is to help ensure the integrity of the IP header during transit through the network.

However, the decision not to include the data payload in the checksum was made for several reasons:

Efficiency: Calculating a checksum over the entire data payload for each IP packet would be computationally expensive and could introduce significant processing overhead on routers and end systems. Given that higher-layer protocols, such as Transmission Control Protocol (TCP) and User Datagram Protocol (UDP), often have their own checksums for data integrity, it was deemed unnecessary to duplicate this effort at the IP layer.

Flexibility: IP is a network layer protocol that operates independently of the transport layer protocols (like TCP or UDP) that carry the actual data. By excluding the data payload from the IP checksum, IP can be used to carry various types of data, regardless of the transport protocol, without modification to the IP layer.

End-to-End Checksums: Higher-layer protocols, especially those in the transport layer (e.g., TCP and UDP), typically include their own checksums that cover both the header and the data payload. This provides end-to-end error detection, ensuring that data integrity is maintained from the source to the destination.

In summary, the decision to include only the IP header in the checksum was a trade-off between efficiency and flexibility. The responsibility for data integrity is often delegated to higher-layer protocols, which can perform more sophisticated error detection and correction based on their specific requirements.

8. Differentiate between Open loop and Closed loop approaches for congestion control. Give one example for each.

S.No.	Open-Loop Control System	Closed-Loop Control System
1.	It is easier to build.	It is difficult to build.
2.	It can perform better if the calibration is properly done.	It can perform better because of the feedback.
3.	It is more stable.	It is comparatively less stable.

S.No.	Open-Loop Control System	Closed-Loop Control System
4.	Optimization for desired output can not be performed.	Optimization can be done very easily.
5.	It does not consists of feedback mechanism.	Feedback mechanism is present.
6.	It requires less maintenance.	Maintenance is difficult.
7.	It is less reliable.	It is more reliable.
8.	It is comparatively slower.	It is faster.
9.	It can be easily installed and is economical.	Complicated installation is required and is expensive.

One example for open loop congestion control

-Acknowledgement policy

One example for closed loop congestion control

-Choke Packets

9. Why is Transport layer called true End to End layer? Why is flow control and error control used in Transport layer in addition to data link layer ?

The Transport layer is often referred to as the "true end-to-end layer" because it provides end-to-end communication and services between processes running on different devices in a network. This layer is responsible for ensuring reliable and efficient communication between applications on separate hosts. The two key reasons why the Transport layer is considered the "true end-to-end layer" are:

End-to-End Communication:

The Transport layer establishes communication between two devices, regardless of the number of intermediate devices (routers, switches, etc.) in the network. It ensures that data is reliably and accurately delivered from the source application to the destination application.

End-to-End Services:

The Transport layer provides end-to-end services such as flow control, error control, and multiplexing/demultiplexing to support reliable and efficient communication between applications on different hosts.

Now, regarding the use of flow control and error control in the Transport layer in addition to the Data Link layer:

Flow Control:

Flow control is a mechanism used to manage the rate of data transmission between sender and receiver to prevent the receiver from being overwhelmed by a fast sender. The Transport layer implements flow control to ensure that a fast sender does not flood a slower receiver with more data than it can handle. Flow control at the Transport layer is end-to-end, considering the characteristics and capabilities of the source and destination devices.

Error Control:

Error control involves detecting and correcting errors that may occur during the transmission of data. While the Data Link layer typically handles error detection and correction within a local network segment, the Transport layer focuses on end-to-end error control. This means that errors occurring across multiple network segments are detected and corrected at the Transport layer to ensure the integrity of the data being exchanged between applications.

In summary, the Transport layer is considered the "true end-to-end layer" because it provides end-to-end communication services between applications on different devices. Flow control and error control mechanisms at the Transport layer are designed to address the unique challenges of end-to-end communication, taking into account the characteristics of the entire communication path, including multiple network segments and devices.

10. aDescribe the ports used by the FTP with suitable diagram.

File Transfer Protocol(FTP) is an application layer protocol that moves files between local and remote file systems. It runs on top of TCP, like HTTP. To transfer a file, 2 TCP connections are used by FTP in parallel: control connection and data connection.

Types of Connection in FTP

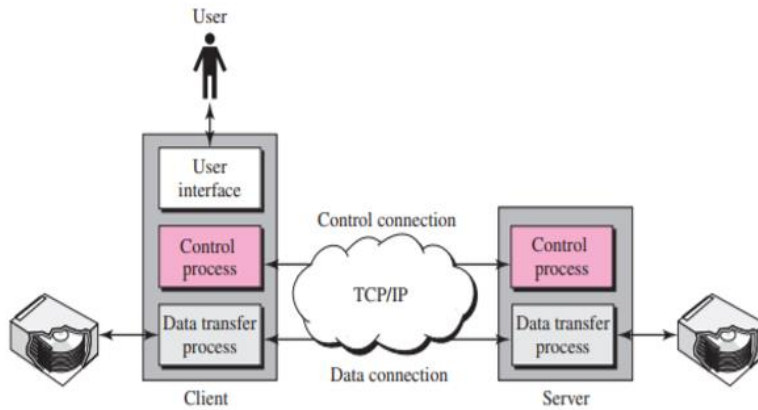
Control Connection

Data Connection

1. Control Connection: For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.

2. Data connection: For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20.

FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.



PART B

- 11 a) Justify why TCP/ IP reference model is called as a protocol stack model. Describe TCP/IP model with suitable diagrams.

The TCP/IP reference model is often referred to as a "protocol stack model" because it organizes networking protocols into a layered structure, with each layer providing specific functions and services. The concept of a stack signifies the hierarchical arrangement of these layers, where each layer builds upon the functionalities provided by the layer beneath it. This layered architecture allows for modularity, ease of implementation, and interoperability between different systems and network devices.

Layers of TCP/IP Model

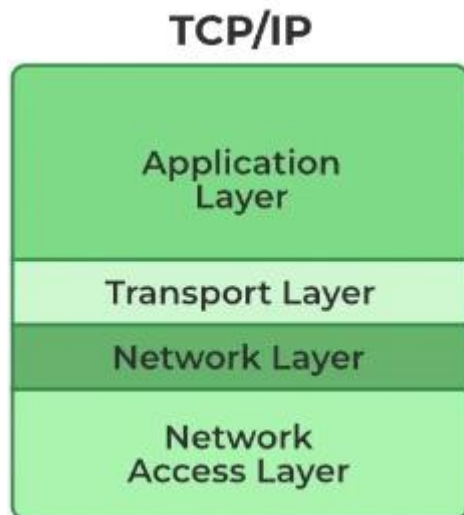
Application Layer

Transport Layer(TCP/UDP)

Network/Internet Layer(IP)

Data Link Layer (MAC)

Physical Layer



1. Physical Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

2. Data Link Layer

The packet's network protocol type, in this case, TCP/IP, is identified by the data-link layer. Error prevention and “framing” are also provided by the data-link layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

3. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

IP: IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.

ICMP: ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.

ARP: ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address,

which is used to identify the device and determine the route that packets should take to reach it.

4. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

TCP: Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.

UDP: The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

5. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

HTTP and HTTPS: HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.

SSH: SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.

NTP: NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

The "stack" terminology emphasizes the idea that these layers are stacked on top of each other, and data flows down through the layers during transmission and up during reception. Each layer performs specific tasks, and the communication between layers is defined by standardized interfaces.

This model is often contrasted with the OSI (Open Systems Interconnection) model, which is another popular reference model. While the OSI model has seven layers, the TCP/IP model, developed before OSI, effectively accomplishes similar objectives with its four layers, making it widely used in practical networking implementations.

11b) Compare LAN, MAN, WAN.

Basis	LAN	MAN	WAN
Full-Form	LAN stands for local area network.	MAN stands for metropolitan area network.	WAN stands for wide area network.
Geographic Span	Operates in small areas such as the same building or campus.	Operates in large areas such as a city.	Operates in larger areas such as country or continent.
Ownership	LAN's ownership is private.	MAN's ownership can be private or public.	While WAN also might not be owned by one organization.
Transmission Speed	The transmission speed of a LAN is high.	While the transmission speed of a MAN is average.	Whereas the transmission speed of a WAN is low.
Propagation delay	The propagation delay is short in a LAN.	There is a moderate propagation delay in a MAN.	Whereas, there is a long propagation delay in a WAN.
Congestion	There is less congestion in LAN.	While there is more congestion in MAN.	Whereas there is more congestion than MAN in WAN.
Design & Maintenance	LAN's design and maintenance are easy.	While MAN's design and maintenance are difficult than LAN.	Whereas WAN's design and maintenance are also difficult than LAN as well MAN.
Fault tolerance	There is more fault tolerance in LAN.	While there is less fault tolerance.	In WAN, there is also less fault tolerance.

12 a) Compare the types of networks formed based on topology. Discuss the advantages and disadvantages of each type.

In Computer Network ,there are various ways through which different components are connected to one another. Network Topology is the way that defines the structure, and how these components are connected to each other.

Types of Network Topology

The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as Network Topology. The various network topologies are:

Point to Point Topology

Mesh Topology

Star Topology

Bus Topology

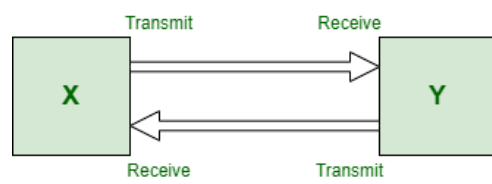
Ring Topology

Tree Topology

Hybrid Topology

Point to Point Topology

Point-to-Point Topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



Mesh Topology

In a mesh topology, every device is connected to another device via a particular channel. In Mesh Topology, the protocols used are AHCP (Ad Hoc Configuration Protocols), DHCP (Dynamic Host Configuration Protocol), etc.

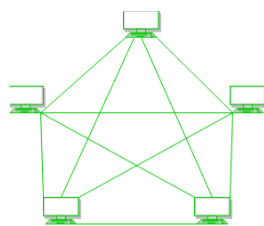


Figure 1: Every device is connected to another via dedicated channels. These channels are known as links.

Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is N-1. In Figure 1, there are 5 devices connected to each other, hence the total number of ports required by each device is 4. The total number of ports required = $N * (N-1)$.

Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is $\frac{N(N-1)}{2}$ i.e. $\frac{N(N-1)}{2}$. In Figure 1, there are 5 devices connected to each other, hence the total number of links required is $\frac{5*4}{2} = 10$.

Advantages of Mesh Topology

Communication is very fast between the nodes.

Mesh Topology is robust.

The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.

Provides security and privacy.

Drawbacks of Mesh Topology

Installation and configuration are difficult.

The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.

The cost of maintenance is high.

A common example of mesh topology is the internet backbone, where various internet service providers are connected to each other via dedicated channels. This topology is also used in military communication systems and aircraft navigation systems.

For more, refer to the Advantages and Disadvantages of Mesh Topology.

Star Topology

In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them. Coaxial cables or RJ-45 cables are used to connect the computers. In Star Topology, many popular Ethernet LAN protocols are used as CD(Collision Detection), CSMA (Carrier Sense Multiple Access), etc.

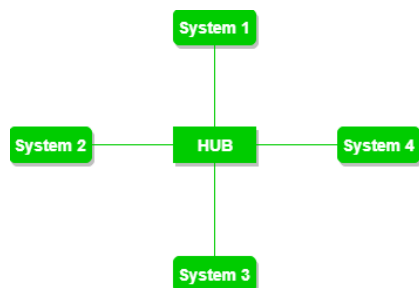


Figure 2: A star topology having four systems connected to a single point of connection i.e. hub.

Advantages of Star Topology

If N devices are connected to each other in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.

Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.

It is Robust. If one link fails only that link will affect and not other than that.

Easy to fault identification and fault isolation.

Star topology is cost-effective as it uses inexpensive coaxial cable.

Drawbacks of Star Topology

If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.

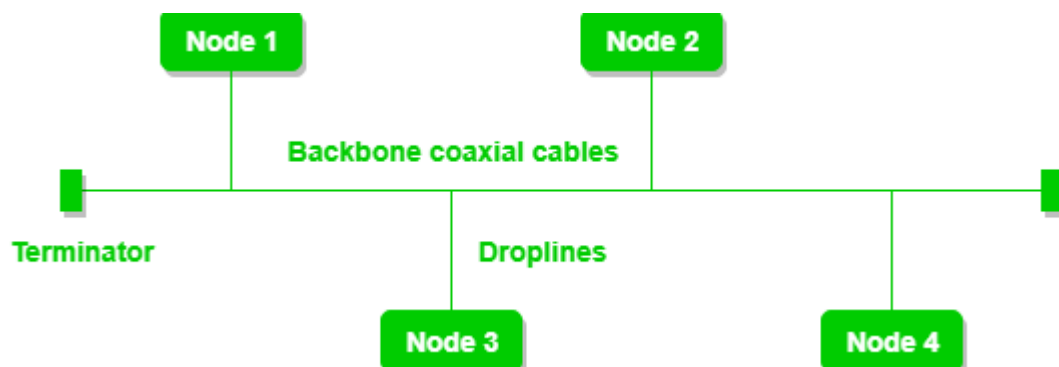
The cost of installation is high.

Performance is based on the single concentrator i.e. hub.

A common example of star topology is a local area network (LAN) in an office where all computers are connected to a central hub. This topology is also used in wireless networks where all devices are connected to a wireless access point.

Bus Topology

Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. In Bus Topology, various MAC (Media Access Control) protocols are followed by LAN ethernet connections like TDMA, Pure Aloha, CDMA, Slotted Aloha, etc.



Bus Topology

Figure 3: A bus topology with shared backbone cable. The nodes are connected to the channel via drop lines.

Advantages of Bus Topology

If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.

Coaxial or twisted pair cables are mainly used in bus-based networks that support up to 10 Mbps.

The cost of the cable is less compared to other topologies, but it is used to build small networks.

Bus topology is familiar technology as installation and troubleshooting techniques are well known.

CSMA is the most common method for this type of topology.

Drawbacks of Bus Topology

A bus topology is quite simpler, but still, it requires a lot of cabling.

If the common cable fails, then the whole system will crash down.

If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD, etc.

Adding new devices to the network would slow down networks.

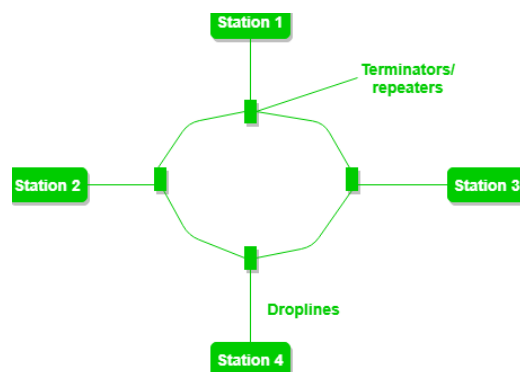
Security is very low.

A common example of bus topology is the Ethernet LAN, where all devices are connected to a single coaxial cable or twisted pair cable. This topology is also used in cable television networks. For more, refer to the Advantages and Disadvantages of Bus Topology.

Ring Topology

In a Ring Topology, it forms a ring connecting devices with exactly two neighboring devices. A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.

The data flows in one direction, i.e. it is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called Dual Ring Topology. In-Ring Topology, the Token Ring Passing protocol is used by the workstations to transmit the data.



Ring Topology

Figure 4: A ring topology comprises 4 stations connected with each forming a ring.

The most common access method of ring topology is token passing.

Token passing: It is a network access method in which a token is passed from one node to another node.

Token: It is a frame that circulates around the network.

Operations of Ring Topology

One station is known as a monitor station which takes all the responsibility for performing the operations.

To transmit the data, the station has to hold the token. After the transmission is done, the token is to be released for other stations to use.

When no station is transmitting the data, then the token will circulate in the ring.

There are two types of token release techniques: Early token release releases the token just after transmitting the data and Delayed token release releases the token after the acknowledgment is received from the receiver.

Advantages of Ring Topology

The data transmission is high-speed.

The possibility of collision is minimum in this type of topology.

Cheap to install and expand.

It is less costly than a star topology.

Drawbacks of Ring Topology

The failure of a single node in the network can cause the entire network to fail.

Troubleshooting is difficult in this topology.

The addition of stations in between or the removal of stations can disturb the whole topology.

Less secure.

For more, refer to the Advantages and Disadvantages of Ring Topology.

Tree Topology

This topology is the variation of the Star topology. This topology has a hierarchical flow of data. In Tree Topology, protocols like DHCP and SAC (Standard Automatic Configuration) are used.

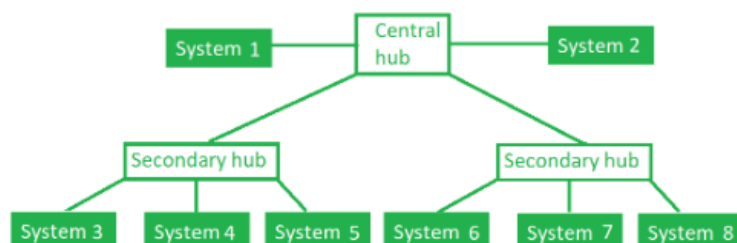


Figure 5: In this, the various secondary hubs are connected to the central hub which contains the repeater. This data flow from top to bottom i.e. from the central hub to the secondary and

then to the devices or from bottom to top i.e. devices to the secondary hub and then to the central hub. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes.

Advantages of Tree Topology

It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.

It allows the network to get isolated and also prioritize from different computers.

We can add new devices to the existing network.

Error detection and error correction are very easy in a tree topology.

Drawbacks of Tree Topology

If the central hub gets fails the entire system fails.

The cost is high because of the cabling.

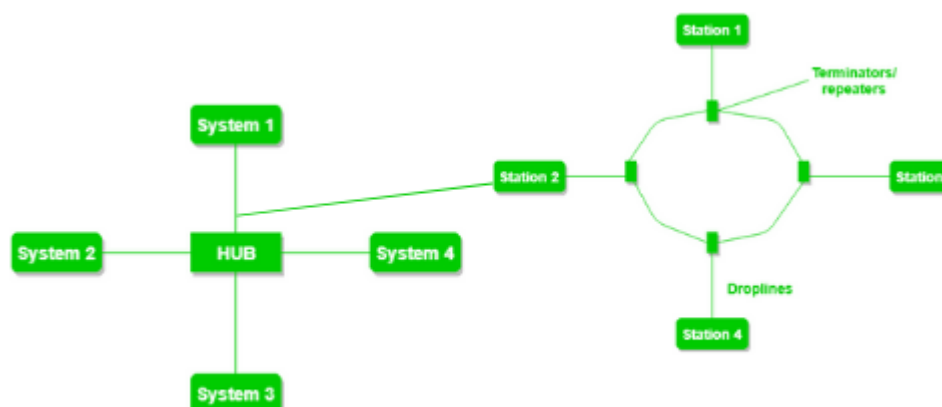
If new devices are added, it becomes difficult to reconfigure.

A common example of a tree topology is the hierarchy in a large organization. At the top of the tree is the CEO, who is connected to the different departments or divisions (child nodes) of the company. Each department has its own hierarchy, with managers overseeing different teams (grandchild nodes). The team members (leaf nodes) are at the bottom of the hierarchy, connected to their respective managers and departments.

For more, refer to the Advantages and Disadvantages of Tree Topology.

Hybrid Topology

This topological technology is the combination of all the various types of topologies we have studied above. Hybrid Topology is used when the nodes are free to take any form. It means these can be individuals such as Ring or Star topology or can be a combination of various types of topologies seen above. Each individual topology uses the protocol that has been discussed earlier.



Hybrid Topology

Figure 6: The above figure shows the structure of the Hybrid topology. As seen it contains a combination of all different types of networks.

Advantages of Hybrid Topology

This topology is very flexible.

The size of the network can be easily expanded by adding new devices.

Drawbacks of Hybrid Topology

It is challenging to design the architecture of the Hybrid Network.

Hubs used in this topology are very expensive.

The infrastructure cost is very high as a hybrid network requires a lot of cabling and network devices.

A common example of a hybrid topology is a university campus network. The network may have a backbone of a star topology, with each building connected to the backbone through a switch or router. Within each building, there may be a bus or ring topology connecting the different rooms and offices. The wireless access points also create a mesh topology for wireless devices. This hybrid topology allows for efficient communication between different buildings while providing flexibility and redundancy within each building.

- 12b) Illustrate the construction of optical fiber and justify how this reduces the interference. Draw the structure of optical fiber.

Optical fiber

- Optical fiber is a cable that accepts and transports signals in the form of light.
- A fiber-optic cable is made of glass or plastic
- It consists of an inner glass core surrounded by a glass cladding which has lower refractive index.
- Optical fibers use reflection to guide light through a channel.
- Light is launched into the fiber using a light source such as light emitting diode(LED) Or laser
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic.

Optical fiber reduces interference in data transmission primarily because it uses light signals to transmit information instead of electrical signals. This characteristic offers several advantages that contribute to a high level of resistance against interference:

Electromagnetic Interference (EMI) Immunity:

Optical fibers transmit data using light signals (typically in the infrared range) rather than electrical signals. As a result, they are not susceptible to electromagnetic interference (EMI) from external sources like radio frequency interference (RFI) or other electronic devices. This immunity to EMI is particularly beneficial in environments with a high density of electronic equipment.

Radio Frequency Interference (RFI) Immunity:

Since optical fibers use light for communication, they are inherently immune to radio frequency interference. Traditional copper-based transmission lines, such as coaxial cables or twisted-pair cables, can be susceptible to RFI, which can degrade signal quality and result in data errors. Optical fibers eliminate this concern.

Cross-talk Minimization:

Cross-talk refers to the unwanted transfer of signals between adjacent cables or channels. In electrical transmission systems, cross-talk can occur due to the electromagnetic fields generated by nearby conductors. Optical fibers, being non-conductive and utilizing light signals, do not produce electromagnetic fields that can cause cross-talk with neighboring fibers. This leads to a significant reduction in cross-talk-related interference.

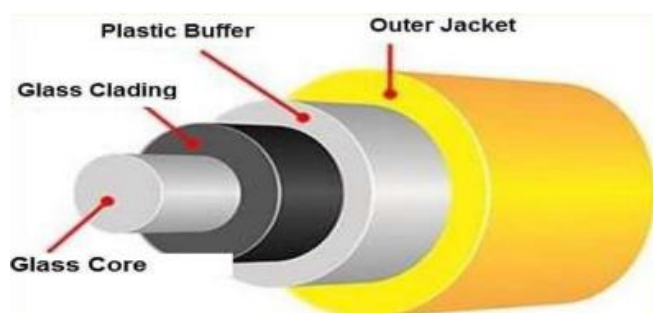
Security against Signal Tapping:

Optical fibers provide a higher level of security against signal tapping or eavesdropping compared to traditional copper cables. Since light signals are contained within the fiber core and do not radiate electromagnetic signals, it is more challenging to intercept or tap the transmitted data without physically accessing the fiber.

Distance without Signal Degradation:

Optical fibers allow for the transmission of signals over much longer distances without significant signal degradation compared to traditional copper-based cables. This property is beneficial in long-distance communication scenarios, as it reduces the need for signal repeaters and amplifiers, which can introduce additional points of vulnerability to interference.

In summary, optical fiber reduces interference by leveraging the characteristics of light signals, which are immune to electromagnetic interference and provide a secure and reliable medium for data transmission. These qualities make optical fiber a preferred choice in environments where interference resilience is crucial, such as telecommunications networks, high-performance computing, and long-distance data transmission applications.



13 a) Summarize about the high-speed LANs specified in IEEE standard.

IEEE has several standards for high-speed Local Area Networks (LANs). One of the most prominent standards is IEEE 802.3, which covers Ethernet LANs. Here's a summary of IEEE 802.3 standards for high-speed LANs:

IEEE 802.3 Ethernet Standards:

802.3ae (10 Gigabit Ethernet): This standard defines 10 Gigabit Ethernet (10GbE), providing higher data rates than previous Ethernet standards. It supports both fiber optic and copper cabling, making it suitable for various network architectures.

802.3ba (40 Gigabit Ethernet and 100 Gigabit Ethernet): This standard specifies 40 Gigabit Ethernet (40GbE) and 100 Gigabit Ethernet (100GbE). It addresses the need for higher data rates to support increasing bandwidth requirements in data centers and high-performance computing environments.

802.3bj (100 Gigabit Ethernet Backplane): This standard focuses on 100GbE for backplane applications, providing specifications for high-speed connections within network devices and systems.

802.3by (25 Gigabit Ethernet): This standard introduces 25 Gigabit Ethernet (25GbE) to meet the growing demand for higher-speed connections, especially in data center environments. It is commonly used for server-to-switch interconnections.

802.3bz (2.5 and 5 Gigabit Ethernet): This standard addresses the need for intermediate data rates between 1 Gigabit Ethernet (1GbE) and 10 Gigabit Ethernet (10GbE). It specifies both 2.5GbE and 5GbE over twisted-pair copper cabling.

Other IEEE 802 Standards:

802.11ac (Wireless LAN): While not part of the IEEE 802.3 Ethernet standards, it's worth mentioning that IEEE 802.11ac defines high-speed wireless LAN (Wi-Fi) technology, offering gigabit-level data rates in wireless networks.

802.11ax (Wi-Fi 6): This standard builds on 802.11ac and focuses on improving efficiency and performance in high-density environments with a large number of connected devices.

These standards play a crucial role in meeting the increasing demand for higher network speeds in various applications, including data centers, enterprise networks, and telecommunications. They address the need for both wired and wireless high-speed LAN solutions to support the growing volume of data traffic and emerging technologies.

13 b) Compare Go back N protocol with Selective Repeat protocol using required diagrams.

Both Go-Back-N Protocol and Selective Repeat Protocol are the types of sliding window protocols. The main difference between these two protocols is that after finding the suspect or damage in sent frames go-back-n protocol re-transmits all the frames whereas selective repeat protocol re-transmits only that frame which is damaged.

Go-Back-N Protocol:

The Go-Back-N protocol is a sliding window protocol used for reliable data transfer in computer networks. It is a sender-based protocol that allows the sender to transmit multiple packets without waiting for an acknowledgement for each packet. The receiver sends a cumulative acknowledgement for a sequence of packets, indicating the last correctly received packet. If any packet is lost, the receiver sends a negative acknowledgement (NACK) for the lost packet, and the sender retransmits all the packets in the window starting from the lost packet. The sender also maintains a timer for each packet, and if an acknowledgement is not received within the timer's timeout period, the sender retransmits all packets in the window.

The key features of the Go-Back-N (GBN) protocol include:

Sliding window mechanism

Sequence numbers

Cumulative acknowledgements

Timeout mechanism

NACK mechanism

Simple implementation.

Selective Repeat Protocol:

The Selective Repeat protocol is another sliding window protocol used for reliable data transfer in computer networks. It is a receiver-based protocol that allows the receiver to acknowledge each packet individually, rather than a cumulative acknowledgement of a sequence of packets. The sender sends packets in a window and waits for acknowledgements for each packet in the window. If a packet is lost, the receiver sends a NACK for the lost packet, and the sender retransmits only that packet. The sender also maintains a timer for each packet, and if an acknowledgement is not received within the timer's timeout period, the sender retransmits only that packet.

key features include:

Receiver-based protocol

Each packet is individually acknowledged by the receiver

Only lost packets are retransmitted, reducing network congestion

Maintains a buffer to store out-of-order packets

Requires more memory and processing power than Go-Back-N

Provides efficient transmission of packets.

Similarities between the two protocols are:

Both protocols use a sliding window mechanism to allow the sender to transmit multiple packets without waiting for an acknowledgement for each packet.

Both protocols use sequence numbers to ensure the correct order of packets.

Both protocols use a timer mechanism to handle lost or corrupted packets.

Both protocols can retransmit packets that are not acknowledged by the receiver.

Both protocols can reduce network congestion by only retransmitting lost packets.

Both protocols are widely used in modern communication networks.

The differences between them are:

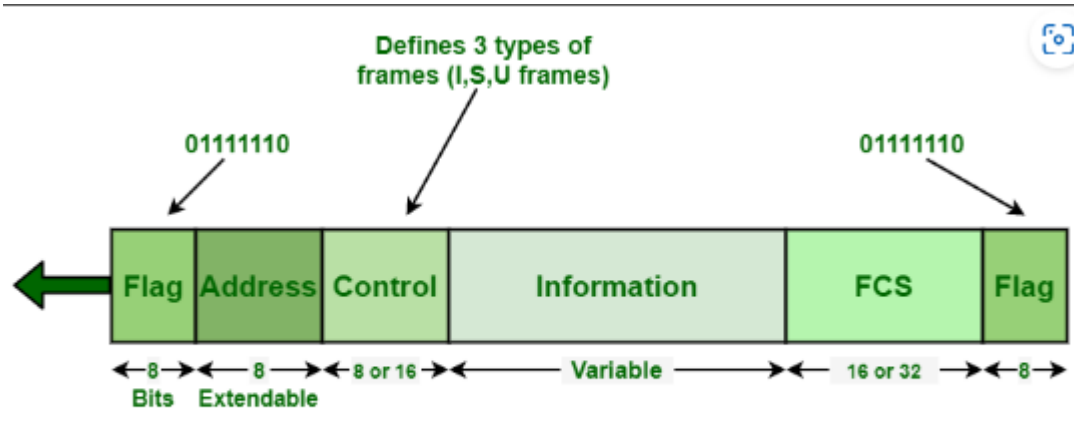
S.NO	Go-Back-N Protocol	Selective Repeat Protocol
1.	In Go-Back-N Protocol , if the sent frame are find suspected then all the frames are re-transmitted from the lost packet to the last packet transmitted.	In selective Repeat protocol , only those frames are re-transmitted which are found suspected.
2.	Sender window size of Go-Back-N Protocol is N.	Sender window size of selective Repeat protocol is also N.
3.	Receiver window size of Go-Back-N Protocol is 1.	Receiver window size of selective Repeat protocol is N.
4.	Go-Back-N Protocol is less complex.	Selective Repeat protocol is more complex.
5.	In Go-Back-N Protocol, neither sender nor at receiver need sorting.	In selective Repeat protocol, receiver side needs sorting to sort the frames.
6.	In Go-Back-N Protocol, type of Acknowledgement is cumulative.	In selective Repeat protocol, type of Acknowledgement is individual.
7.	In Go-Back-N Protocol, Out-of-Order packets are NOT Accepted (discarded) and the entire window is re-transmitted.	In selective Repeat protocol, Out-of-Order packets are Accepted.
8.	In Go-Back-N Protocol, if Receives a corrupt packet, then also, the entire window is re-transmitted.	In selective Repeat protocol, if Receives a corrupt packet, it immediately sends a negative acknowledgement and hence only the selective packet is retransmitted.
9.	Efficiency of Go-Back-N Protocol is $N/(1+2*a)$	Efficiency of selective Repeat protocol is also $N/(1+2*a)$

For fig: refer notes given.

14 a) Discuss about the frame formats of HDLC protocol.

High-Level Data Link Control (HDLC) generally uses term “frame” to indicate and represent an entity of data or a protocol of data unit often transmitted or transferred from one station to another station. Each and every frame on link should begin and end with Flag Sequence Field (F). Each of frames in HDLC includes mainly six fields. It begins with a flag field, an address field, a control field, an information field, an frame check sequence (FCS) field, and an ending flag field. The ending flag field of one frame can serve as beginning flag field of the next frame in multiple-frame transmissions.

The basic frame structure of HDLC protocol is shown below :



Basic Frame Structure

Flag Field –

The flag field is generally responsible for initiation and termination of error checking. In HDLC protocol, there is no start and stop bits. So, the flag field is basically using delimiter 0x7e to simply indicate beginning and end of frame.

It is an 8-bit sequence with a bit pattern 01111110 that basically helps in identifying both starting and end of a frame. This bit pattern also serves as a synchronization pattern for receiver. This bit pattern is also not allowed to occur anywhere else inside a complete frame.

Address Field –

The address field generally includes HDLC address of secondary station. It helps to identify secondary station will sent or receive data frame. This field also generally consists of 8 bits therefore it is capable of addressing 256 addresses. This field can be of 1 byte or several bytes long, it depends upon requirements of network. Each byte can identify up to 128 stations.

This address might include a particular address, a group address, or a broadcast address. A primary address can either be a source of communication or a destination that eliminates requirement of including address of primary.

Control Field –

HDLC generally uses this field to determine how to control process of communication. The control field is different for different types of frames in HDLC protocol. The types of frames

can be Information frame (I-frame), Supervisory frame (S-frame), and Unnumbered frame (U-frame).

This field is a 1-2-byte segment of frame generally requires for flow and error control. This field basically consists of 8 bits but it can be extended to 16 bits. In this field, interpretation of bits usually depends upon the type of frame.

Information Field –

This field usually contains data or information of users sender is transmitting to receiver in an I-frame and network layer or management information in U-frame. It also consists of user's data and is fully transparent. The length of this field might vary from one network to another network.

Information field is not always present in an HDLC frame.

Frame Check Sequence (FCS) –

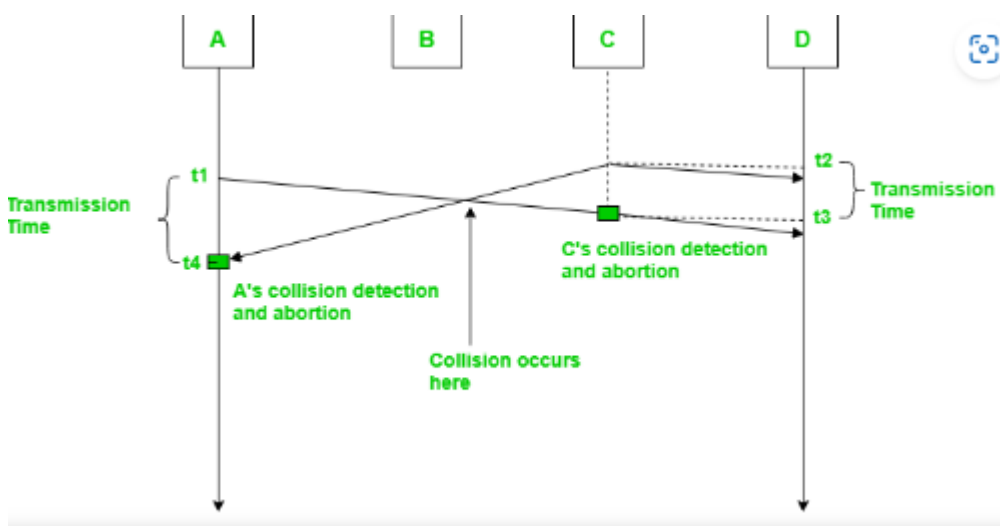
FCS is generally used for identification of errors i.e., HDLC error detection. In FCS, CRC16 (16-bit Cyclic Redundancy Check) or CRC32 (32-bit Cyclic Redundancy Check) code is basically used for error detection. CRC calculation is done again in receiver. If somehow result differs even slightly from value in original frame, an error is assumed.

This field can either contain 2 byte or 4 bytes. This field is a total 16 bit that is required for error detection in address field, control field, and information field. FCS is basically calculated by sender and receiver both of a data frame. FCS is used to confirm and ensure that data frame was not corrupted by medium that is used to transfer frame from sender to receiver.

14 b) Write short notes on any two carrier sense multiple access protocols used in IEEE standards.

1. Carrier Sense Multiple Access with Collision Detection (CSMA/CD):

In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If successful, the transmission is finished, if not, the frame is sent again.



In the diagram, starts sending the first bit of its frame at t_1 and since C sees the channel idle at t_2 , starts sending its frame at t_2 . C detects A's frame at t_3 and aborts transmission. A detects C's frame at t_4 and aborts its transmission. Transmission time for C's frame is, therefore, $t_3 - t_2$ and for A's frame is $t_4 - t_1$

So, the frame transmission time (T_{fr}) should be at least twice the maximum propagation time (T_p). This can be deduced when the two stations involved in a collision are a maximum distance apart.

Throughput and Efficiency: The throughput of CSMA/CD is much greater than pure or slotted ALOHA.

For the 1-persistent method, throughput is 50% when $G=1$.

For the non-persistent method, throughput can go up to 90%.

2. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) –

The basic idea behind CSMA/CA is that the station should be able to receive while transmitting to detect a collision from different stations. In wired networks, if a collision has occurred then the energy of the received signal almost doubles, and the station can sense the possibility of collision. In the case of wireless networks, most of the energy is used for transmission, and the energy of the received signal increases by only 5-10% if a collision occurs. It can't be used by the station to sense collision. Therefore CSMA/CA has been specially designed for wireless networks.

These are three types of strategies:

InterFrame Space (IFS): When a station finds the channel busy it senses the channel again, when the station finds a channel to be idle it waits for a period of time called IFS time. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority.

Contention Window: It is the amount of time divided into slots. A station that is ready to send frames chooses a random number of slots as wait time.

Acknowledgments: The positive acknowledgments and time-out timer can help guarantee a successful transmission of the frame.

Characteristics of CSMA/CA :

Carrier Sense: The device listens to the channel before transmitting, to ensure that it is not currently in use by another device.

Multiple Access: Multiple devices share the same channel and can transmit simultaneously.

Collision Avoidance: If two or more devices attempt to transmit at the same time, a collision occurs. CSMA/CA uses random backoff time intervals to avoid collisions.

Acknowledgment (ACK): After successful transmission, the receiving device sends an ACK to confirm receipt.

Fairness: The protocol ensures that all devices have equal access to the channel and no single device monopolizes it.

Binary Exponential Backoff: If a collision occurs, the device waits for a random period of time before attempting to retransmit. The backoff time increases exponentially with each retransmission attempt.

Interframe Spacing: The protocol requires a minimum amount of time between transmissions to allow the channel to be clear and reduce the likelihood of collisions.

RTS/CTS Handshake: In some implementations, a Request-To-Send (RTS) and Clear-To-Send (CTS) handshake is used to reserve the channel before transmission. This reduces the chance of collisions and increases efficiency.

Wireless Network Quality: The performance of CSMA/CA is greatly influenced by the quality of the wireless network, such as the strength of the signal, interference, and network congestion.

Adaptive Behavior: CSMA/CA can dynamically adjust its behavior in response to changes in network conditions, ensuring the efficient use of the channel and avoiding congestion.

Overall, CSMA/CA balances the need for efficient use of the shared channel with the need to avoid collisions, leading to reliable and fair communication in a wireless network.

Types of CSMA Access Modes:

There are 4 types of access modes available in CSMA. It is also referred as 4 different types of CSMA protocols which decide the time to start sending data across shared media.

1-Persistent: It senses the shared channel first and delivers the data right away if the channel is idle. If not, it must wait and continuously track for the channel to become idle and then broadcast the frame without condition as soon as it does. It is an aggressive transmission algorithm.

Non-Persistent: It first assesses the channel before transmitting data; if the channel is idle, the node transmits data right away. If not, the station must wait for an arbitrary amount of time (not continuously), and when it discovers the channel is empty, it sends the frames.

P-Persistent: It consists of the 1-Persistent and Non-Persistent modes combined. Each node observes the channel in the 1-Persistent mode, and if the channel is idle, it sends a frame with a P probability. If the data is not transferred, the frame restarts with the following time slot after waiting for a $(q = 1-p)$ probability random period.

O-Persistent: A supervisory node gives each node a transmission order. Nodes wait for their time slot according to their allocated transmission sequence when the transmission medium is idle.

Advantages of CSMA:

Increased efficiency: CSMA ensures that only one device communicates on the network at a time, reducing collisions and improving network efficiency.

Simplicity: CSMA is a simple protocol that is easy to implement and does not require complex hardware or software.

Flexibility: CSMA is a flexible protocol that can be used in a wide range of network environments, including wired and wireless networks.

Low cost: CSMA does not require expensive hardware or software, making it a cost-effective solution for network communication.

Disadvantages of CSMA:

Limited scalability: CSMA is not a scalable protocol and can become inefficient as the number of devices on the network increases.

Delay: In busy networks, the requirement to sense the medium and wait for an available channel can result in delays and increased latency.

Limited reliability: CSMA can be affected by interference, noise, and other factors, resulting in unreliable communication.

Vulnerability to attacks: CSMA can be vulnerable to certain types of attacks, such as jamming and denial-of-service attacks, which can disrupt network communication.

15 a) Compare the features of link state routing with distance vector routing.

Distance Vector Routing –

It is a dynamic routing algorithm in which each router computes a distance between itself and each possible destination i.e. its immediate neighbors.

The router shares its knowledge about the whole network to its neighbors and accordingly updates the table based on its neighbors.

The sharing of information with the neighbors takes place at regular intervals.

It makes use of Bellman-Ford Algorithm for making routing tables.

Problems – Count to infinity problem which can be solved by splitting horizon.

– Good news spread fast and bad news spread slowly.

– Persistent looping problem i.e. loop will be there forever.

Link State Routing –

It is a dynamic routing algorithm in which each router shares knowledge of its neighbors with every other router in the network.

A router sends its information about its neighbors only to all the routers through flooding.

Information sharing takes place only whenever there is a change.

It makes use of Dijkstra's Algorithm for making routing tables.

Problems – Heavy traffic due to flooding of packets.

– Flooding can result in infinite looping which can be solved by using the Time to live (TTL) field.

S.No.	Distance Vector Routing	Link State Routing
1.	Bandwidth required is less due to local sharing, small packets and no flooding.	Bandwidth required is more due to flooding and sending of large link state packets.
2.	Based on local knowledge, since it updates table based on information from neighbours.	Based on global knowledge, it have knowledge about entire network.

S.No.	Distance Vector Routing	Link State Routing
3.	Make use of Bellman Ford Algorithm.	Make use of Dijkstra's algorithm.
4.	Traffic is less.	Traffic is more.
5.	Converges slowly i.e, good news spread fast and bad news spread slowly.	Converges faster.
6.	Count of infinity problem.	No count of infinity problem.
7.	Persistent looping problem i.e, loop will be there forever.	No persistent loops, only transient loops.
8.	Practical implementation is RIP and IGRP.	Practical implementation is OSPF and ISIS.

15 b) Differentiate between the implementation of datagram subnet and virtual circuit subnet.

Both Virtual Circuits and Datagram Networks are the types of connection services which are used for transmission of information from a sender to a receiver. The most basic difference between the two is that the virtual circuits are connection oriented services that require resources like buffers, CPU, bandwidth, etc., for a data transfer session, while the datagram networks are connectionless services where no such resources are required for data transmission.

Virtual Circuit is a connection oriented service in which resources like buffers, CPU, bandwidth, etc. are used for creating a data transfer session. Virtual Circuit is also known as connection-oriented switching.

In virtual circuits, the path that is followed by the first data packet would get fixed and all other data packets will also use the same path and consume the same resources. Consequently, a common and same header is used by all the data packets.

Due to all these reasons, virtual circuits are comparatively less complex and more reliable for data transmission, however they are expensive to install and maintain. Virtual circuits are mostly used in Asynchronous Transfer Mode (ATM) Networks that are used for the telephone calls.

Datagram networks are connectionless services for data transmission, in which no resources like CPU, buffer, bandwidth, etc. are required for data transmission. In datagram networks, the path for data transmission is not fixed. Therefore, the data packets are free to decide the path on any intermediate router on the way by dynamically changing the routing tables on the routers.

Since the data packets follow different paths, they have different headers with information of the data packets. Due to dynamic resource allocation and dynamic path, datagram networks are errorprone and less reliable. However, datagram networks are cheaper to install and maintain. They are widely used in IP (Internet Protocol) services like internet.

Key	Virtual Circuits	Datagram Networks
Definition	Virtual Circuit is a connection-oriented service in which there is an implementation of resources like buffers, CPU, bandwidth, etc., used by virtual circuit for a data transfer session.	Datagram networks are a type of connectionless service where no such resources are required for data transmission.
Path	In Virtual circuits, as all the resources and bandwidth get reserved before the transmission, the path which is utilized or followed by first data packet would get fixed and all other data packets will use the same path and consume same resources.	In a Datagram network, the path is not fixed as data packets are free to decide the path on any intermediate router on the go by dynamically changing routing tables on routers.
Header	As there is same path followed by all the data packets, a common and same header is being used by all the packets.	Different headers with information of other data packet is being used in Datagram network.
Complexity	Virtual Circuit is less complex as compared to that of Datagram network.	Datagram network are more complex as compared to Virtual circuit.
Reliability	Due to fixed path and assurance of fixed resources, Virtual Circuits are more reliable for data transmission as compared to Datagram network.	Datagram networks, due to their dynamic resource allocation and dynamic path, are more errorprone and less reliable than Virtual circuits.

Example
and Cost

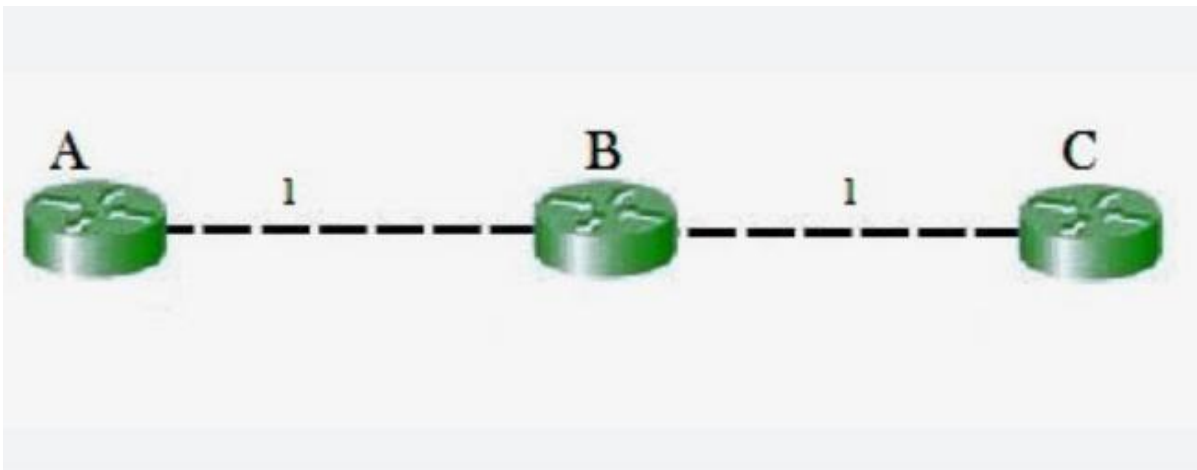
Virtual circuits are costlier in installation and maintenance. They are widely used by ATM (Asynchronous Transfer Mode) Network, which is used for the Telephone calls.

Datagram networks are cheaper as compared to the Virtual Circuits. They are mainly used by IP network, which is used for Data services like Internet.

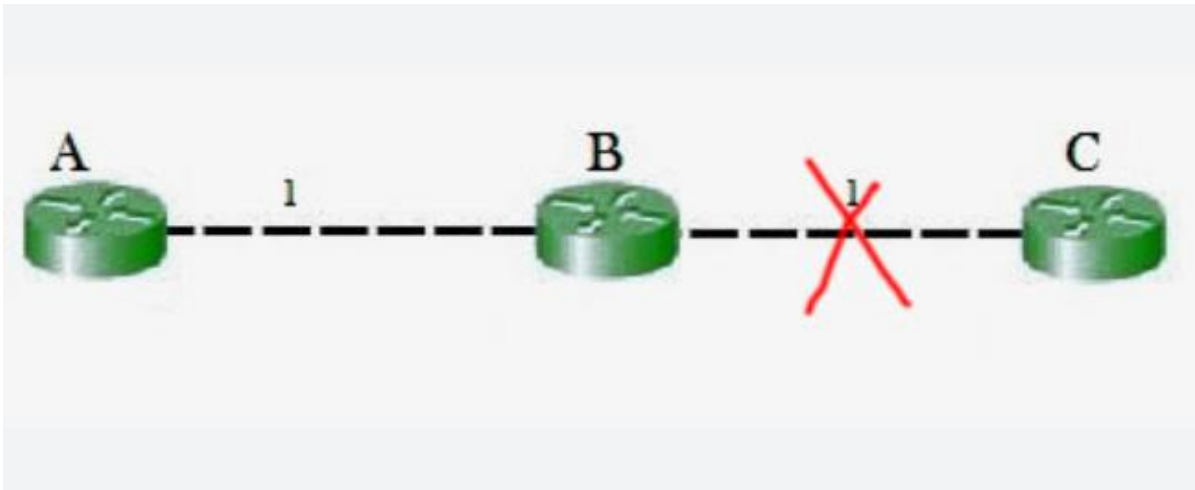
16 a) What is count to infinity problem? Describe two techniques to solve the count to infinity problem in distance vector routing algorithm.

The main issue with Distance Vector Routing (DVR) protocols is Routing Loops since Bellman-Ford Algorithm cannot prevent loops. This routing loop in the DVR network causes the Count to Infinity Problem. Routing loops usually occur when an interface goes down or two routers send updates at the same time.

Counting to infinity problem:



So in this example, the Bellman-Ford algorithm will converge for each router, they will have entries for each other. B will know that it can get to C at a cost of 1, and A will know that it can get to C via B at a cost of 2.



If the link between B and C is disconnected, then B will know that it can no longer get to C via that link and will remove it from its table. Before it can send any updates it's possible that it will receive an update from A which will be advertising that it can get to C at a cost of 2. B can get to A at a cost of 1, so it will update a route to C via A at a cost of 3. A will then receive updates from B later and update its cost to 4. They will then go on feeding each other bad information toward infinity which is called as Count to Infinity problem.

Solutions to Count to infinity problem:

Split Horizon

The split horizon is a method for resolving instability. Each node in this technique delivers a portion of its table across each interface rather than flooding the table through all of them. If node B believes that the best way to go to X is through node A, then node B need not inform node A of this information because it has already been provided by node A (A already knows).

Confusion is brought on by receiving data from node A, altering it, then sending it back to node A. In our example, node B trims the end of its forwarding table before transmitting it to node A. In this instance, node A maintains a distance to X of infinity. Later, node B similarly makes corrections to its forwarding table when node A transmits it to it. After the initial update, the system is stable because both nodes A and B are aware that X cannot be reached.

Poison Reverse

The split-horizon approach has one disadvantage. If there is no news about a route after a certain amount of time, the corresponding protocol typically employs a timer and instructs the node to remove the route from its table. In the previous example, node A is unable to determine whether node B's decision to remove the route to X from its advertisement to node A is the result of the split-horizon technique (the information came from A) or the fact that B has not recently received any news concerning X.

In the poison reverse technique, B can still state the value for X, but if the information came from A, it can substitute infinite for the distance as a warning: " Do not utilise this value; you are the source of my knowledge regarding this route."

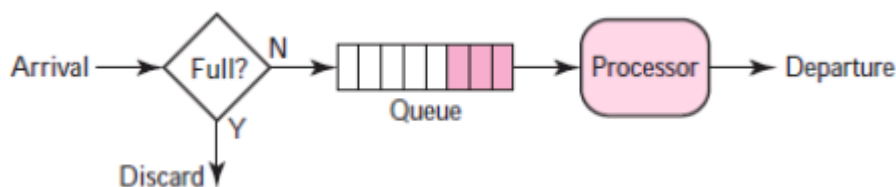
16 b) Identify and describe the scheduling techniques to improve the Quality of Services (QoS)

Techniques that can be used to improve the quality of service as follows scheduling, traffic shaping, admission control and resource reservation.

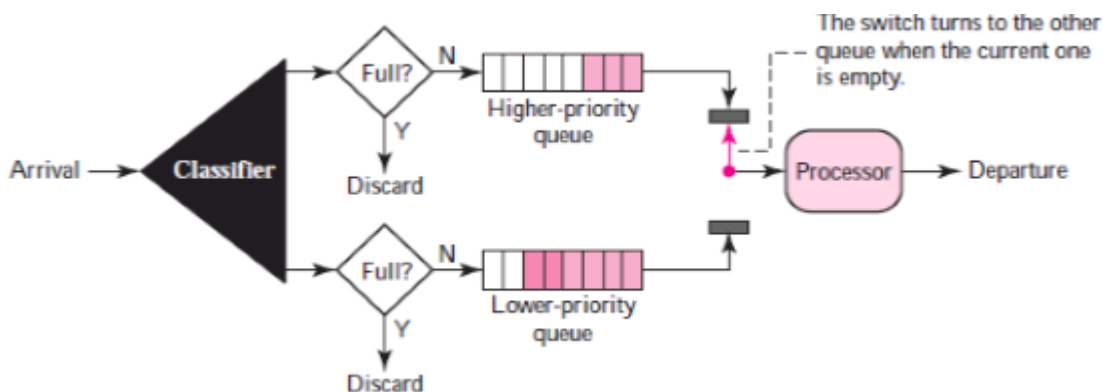
- Scheduling :

Packets from different flows arrive at a switch or router for processing. A good scheduling technique treats the different flows in a fair and appropriate manner. Several scheduling techniques are designed to improve the quality of service. Three of them here: FIFO queuing, priority queuing, and weighted fair queuing.

FIFO Queuing: In first-in, first-out (FIFO) queuing, packets wait in a buffer (queue) until the node (router or switch) is ready to process them. If the average arrival rate is higher than the average processing rate, the queue will fill up and new packets will be discarded. Figure1 shows a conceptual view of a FIFO queue.



Priority Queuing: In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue. The packets in the highest-priority queue are processed first. Packets in the lowest-priority queue are processed last. Note that the system does not stop serving a queue until it is empty. Figure2 shows priority queuing with two priority levels (for simplicity).



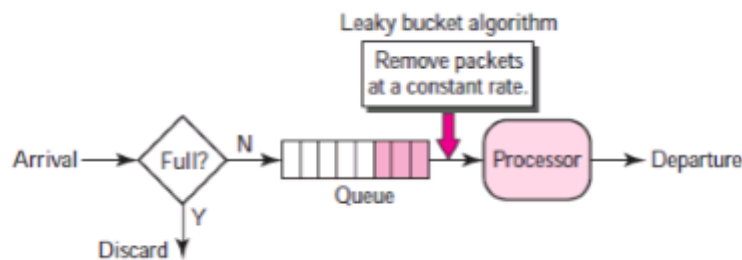
A priority queue can provide better QoS than the FIFO queue because higher priority traffic, such as multimedia, can reach the destination with less delay.

Weighted Fair Queuing: A better scheduling method is weighted fair queuing. In this technique, the packets are still assigned to different classes and admitted to different queues. The queues, however, are weighted based on the priority of the queues; higher priority means a higher weight. The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on the corresponding weight.

Traffic Shaping : Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network. Two techniques can shape traffic: leaky bucket and token bucket.

Leaky Bucket: A technique called leaky bucket can smooth out bursty traffic. Bursty chunks are stored in the bucket and sent out at an average rate.

A simple leaky bucket implementation is shown in Figure3. A FIFO queue holds the packets. If the traffic consists of fixed-size packets, the process removes a fixed number of packets from the queue at each tick of the clock. If the traffic consists of variable-length packets, the fixed output rate must be based on the number of bytes or bits.



The following is an algorithm for variable-length packets:

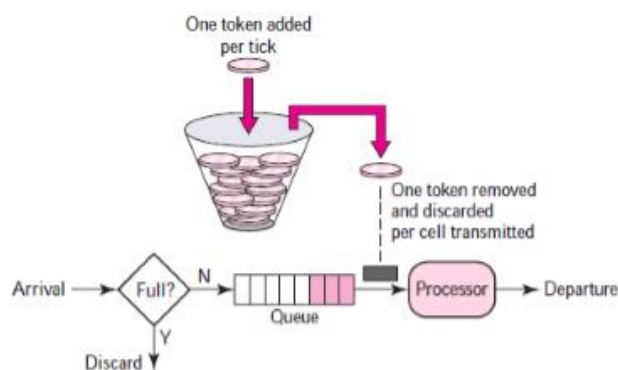
Initialize a counter to n at the tick of the clock.

If n is greater than the size of the packet, send the packet and decrement the counter by the packet size. Repeat this step until n is smaller than the packet size.

Reset the counter and go to step 1.

Token Bucket: The token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens. For each tick of the clock, the system sends n tokens to the bucket. The system removes one token for every cell (or byte) of data sent. For example, if n is 100 and the host is idle for 100 ticks, the bucket collects 10,000 tokens. Now the host can consume all these tokens in one tick with 10,000 cells, or the host takes 1,000 ticks with 10 cells per tick. In other words, the host can send bursty data as long as the bucket is not empty.

Figure 4 shows the idea.



Resource Reservation :

A flow of data needs resources such as a buffer, bandwidth, CPU time, and so on. The quality of service is improved if these resources are reserved beforehand.

Admission Control :

Admission control refers to the mechanism used by a router, or a switch, to accept or reject a flow based on predefined parameters called flow specifications. Before a router accepts a flow for processing, it checks the flow specifications to see if its capacity (in terms of bandwidth, buffer size, CPU speed, etc.) and its previous commitments to other flows can handle the new flow.

- 17 a) Illustrate the sub-netting concept. A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Find the number of subnets possible and hosts which can be connected in each subnet.

Default subnet mask for class B= 255.255.0.0

For 1000 subnets, 10 bits are needed. So 10 bits should be borrowed from host portion to subnet portion.

Then the subnet mask will be 255.255.255.192

Number of subnets = 2^{10}

$$= 1024$$

Number of hosts that can be connected in each subnet = $2^6 - 2$

$$= 62$$

- 17 b) Identify the characteristics of BGP.

Border Gateway Protocol (BGP) is used to Exchange routing information for the internet and is the protocol used between ISP which are different ASes.

The protocol can connect together any internetwork of autonomous system using an arbitrary topology. The only requirement is that each AS have at least one router that is able to run BGP and that is router connect to at least one other AS's BGP router. BGP's main function is to exchange network reach-ability information with other BGP systems. Border Gateway Protocol constructs an autonomous systems' graph based on the information exchanged between BGP routers.

Characteristics of Border Gateway Protocol (BGP):

Inter-Autonomous System Configuration: The main role of BGP is to provide communication between two autonomous systems.

BGP supports Next-Hop Paradigm.

Coordination among multiple BGP speakers within the AS (Autonomous System).

Path Information: BGP advertisement also include path information, along with the reachable destination and next destination pair.

Policy Support: BGP can implement policies that can be configured by the administrator. For ex:- a router running BGP can be configured to distinguish between the routes that are known within the AS and that which are known from outside the AS.

Runs Over TCP.

BGP conserve network Bandwidth.

BGP supports CIDR.

BGP also supports Security.

18 a) Describe how does OSPF perform routing in larger networks? Also explain the different types of OSPF messages.

The OSPF protocol is a link-state routing protocol, which means that the routers exchange topology information with their nearest neighbors. The topology information is flooded throughout the AS, so that every router within the AS has a complete picture of the topology of the AS.

OSPF works in three main steps:

- 1.The first step is to become neighbors with other routers connected to the same segment.
- 2.Then the routers exchange the Link State Advertisements (LSAs) with neighbor routers.
- 3.After that, each router independently calculates the best routes to each destination.

OSPF uses a cost metric that represents the status of the link and the bandwidth of the interface in an algorithm to determine the best route to a destination. The algorithm used is called the SPF (shortest path first) or Dijkstra algorithm.

There are 5 different messages which are used in the OSPF process. They are:

Hello

DBD (Database Description)

LSR (Link State Request)

LSU (Link State Update)

LS-Ack (Link State Acknowledgement)

Hello

Hello, the message is also known as the Type-1 packet. In OSPF, Hello Message is used for:

Neighbor Discovery

Keep Alive (by default timer is 10 seconds)

DBD (Database Description)

DBD stands for Database Description. This message is used to synchronize LSDB (Link State Database) between two Routers. Along with this, this message helps in neighbor formation and MTU size negotiation. This message usually used in the Ex-Start state.

LSR (Link State Request)

LSR stands for Link State Request. It is used to request specific link-state records from an OSPF neighbor router. However, this is initiated by the OSPF Master which is chosen in Ex-Start state.

LSU (Link State Update)

LSU (Link State Update), is a reply to LSR (Link State Request). When the OPSF Master router, sends LSR to slave router, then Slave Router provides Link State Records to Master Router using LSU (Link State Update) message.

LS-Ack (Link State Acknowledgement)

As the name suggests, the LS-Ack message is used for acknowledgment purposes. This acknowledgment is for LSU message. With the help of this message, we can examine that the receiver end OSPF router successfully receives LSU (Link State Update).

18 b) Explain the purposes of using ARP and RARP in the network layer. Also describe the working of each.

In Address Resolution Protocol (ARP), Receiver's MAC address is fetched. Through ARP, (32-bit) IP address mapped into (48-bit) MAC address. Whereas, In Reverse Address Resolution Protocol (RARP), IP address is fetched through server. Through RARP, (48-bit) MAC address of 48 bits mapped into (32-bit) IP address.

Working of ARP

- At the network layer, when the source wants to communicate with the destination. Firstly, the source needs to find out the MAC address (Physical Address) of the destination. For this, the source will check the ARP cache and ARP table for the MAC address of the destination. If the MAC address of the destination is present in the ARP cache or ARP table, then the source uses that MAC address for the communication.
- If the MAC address of the destination is not in the ARP cache or ARP table, then the Source generates an ARP Request message. The ARP Request message consists of the MAC address and the IP address of the source. It also contains the IP address and MAC address of the destination. The MAC address of the destination left null because the user has requested this.

- The ARP Request message will be broadcasted to the local network by the source computer. All the devices in the LAN network receive the broadcast message. Now, each device compares its own IP address with the IP address of the destination. If the IP address of the device match with the IP address of the destination, then that device will send an ARP to reply message. If the IP address of the device does not match the IP address of the destination, then the device will automatically drop the packet.
- The destination sends an ARP reply packet when the destination address matches the device. That ARP Reply packet consists of the MAC address of the device. The destination device automatically updates the table and stores the source's MAC address because this address will be required for the communication from the source.
- Now the source acts as a target for the destination device, and the destination device sends the ARP Reply message.
- The ARP Reply message is unicast instead of broadcast. This is because the device (destination) that is sending the ARP Reply message knows the MAC address of the device (source) to which the ARP Reply message is sent.
- When the source device receives the ARP Reply message, then it will know the MAC address of the destination because the ARP Reply packet contains the MAC address of the destination along with the other addresses. The source will update the MAC address of the destination in the ARP cache. Now the sender is able to communicate directly to the destination.

Advantages of using ARP

- We can easily find out the MAC address of the device if we know the IP address of that device.
- It is not necessary to configure the address of the end nodes for the MAC address. We can find it when needed.

Disadvantages of using ARP

- ARP attacks such as ARP spoofing and ARP denial of service may occur.

Working of RARP

The RARP is on the Network Access Layer and is employed to send data between two points in a very network.

Each network participant has two unique addresses:- IP address (a logical address) and MAC address (the physical address).

The IP address gets assigned by software and after that the MAC address is constructed into the hardware.

The RARP server that responds to RARP requests, can even be any normal computer within the network. However, it must hold the data of all the MAC addresses with their assigned IP addresses. If a RARP request is received by the network, only these RARP servers can reply to it. The info packet needs to be sent on very cheap layers of the network. This implies that the packet is transferred to all the participants at the identical time.

The client broadcasts a RARP request with an Ethernet broadcast address and with its own physical address. The server responds by informing the client its IP address.

19 a) Describe the TCP congestion control approaches with necessary diagrams.

TCP congestion control is a method used by the TCP protocol to manage data flow over a network and prevent congestion. TCP uses a congestion window and congestion policy that avoids congestion. Previously, we assumed that only the receiver could dictate the sender's window size. We ignored another entity here, the network. If the network cannot deliver the data as fast as it is created by the sender, it must tell the sender to slow down. In other words, in addition to the receiver, the network is a second entity that determines the size of the sender's window

Congestion Policy in TCP

Slow Start Phase: Starts slow increment is exponential to the threshold.

Congestion Avoidance Phase: After reaching the threshold increment is by 1.

Congestion Detection Phase: The sender goes back to the Slow start phase or the Congestion avoidance phase.

Slow Start Phase

Exponential increment: In this phase after every RTT the congestion window size increments exponentially.

Example:- If the initial congestion window size is 1 segment, and the first segment is successfully acknowledged, the congestion window size becomes 2 segments. If the next transmission is also acknowledged, the congestion window size doubles to 4 segments. This exponential growth continues as long as all segments are successfully acknowledged.

Initially $cwnd = 1$

After 1 RTT, $cwnd = 2^{(1)} = 2$

2 RTT, $cwnd = 2^{(2)} = 4$

3 RTT, $cwnd = 2^{(3)} = 8$

Congestion Avoidance Phase

Additive increment: This phase starts after the threshold value also denoted as $ssthresh$. The size of $cwnd$ (congestion window) increases additive. After each RTT $cwnd = cwnd + 1$.

Example:- if the congestion window size is 20 segments and all 20 segments are successfully acknowledged within an RTT, the congestion window size would be increased to 21 segments in the next RTT. If all 21 segments are again successfully acknowledged, the congestion window size would be increased to 22 segments, and so on.

Initially $cwnd = i$

After 1 RTT, $cwnd = i+1$

2 RTT, $cwnd = i+2$

3 RTT, $cwnd = i+3$

Congestion Detection Phase

Multiplicative decrement: If congestion occurs, the congestion window size is decreased. The only way a sender can guess that congestion has happened is the need to retransmit a segment. Retransmission is needed to recover a missing packet that is assumed to have been dropped by a router due to congestion. Retransmission can occur in one of two cases: when the RTO timer times out or when three duplicate ACKs are received.

Case 1: Retransmission due to Timeout – In this case, the congestion possibility is high.

(a) $ssthresh$ is reduced to half of the current window size.

(b) set $cwnd = 1$

(c) start with the slow start phase again.

Case 2: Retransmission due to 3 Acknowledgement Duplicates – The congestion possibility is less.

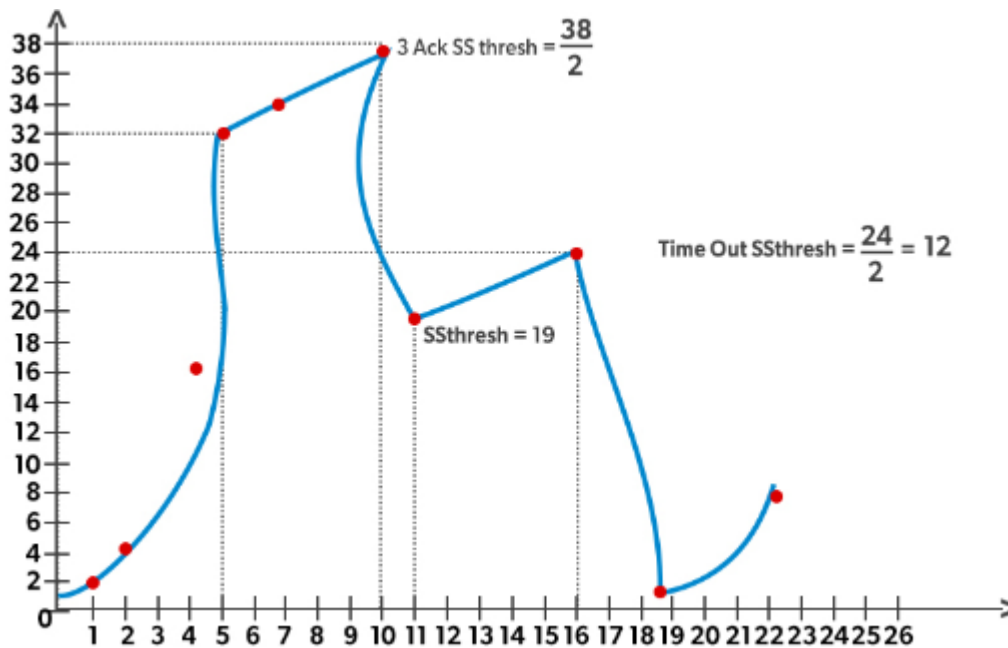
(a) $ssthresh$ value reduces to half of the current window size.

(b) set $cwnd = ssthresh$

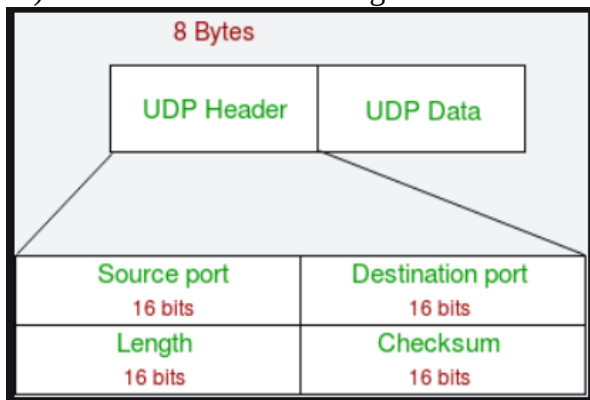
(c) start with congestion avoidance phase

Example

Assume a TCP protocol experiencing the behavior of slow start. At the 5th transmission round with a threshold ($ssthresh$) value of 32 goes into the congestion avoidance phase and continues till the 10th transmission. At the 10th transmission round, 3 duplicate ACKs are received by the receiver and entered into additive increase mode. Timeout occurs at the 16th transmission round. Plot the transmission round (time) vs congestion window size of TCP segments.



19 b) Demonstrate the UDP segment structure.



Source Port: Source Port is a 2 Byte long field used to identify the port number of the source.

Destination Port: It is a 2 Byte long field, used to identify the port of the destined packet.

Length: Length is the length of UDP including the header and the data. It is a 16-bits field.

Checksum: Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, the pseudo-header of information from the IP header, and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.

20 a) Summarize the architecture of electronic mail system with neat diagram.

Electronic mail, commonly known as email, is a method of exchanging messages over the internet. Here are the basics of email:

An email address: This is a unique identifier for each user, typically in the format of name@domain.com.

An email client: This is a software program used to send, receive and manage emails, such as Gmail, Outlook, or Apple Mail.

An email server: This is a computer system responsible for storing and forwarding emails to their intended recipients.

To send an email:

Compose a new message in your email client.

Enter the recipient's email address in the "To" field.

Add a subject line to summarize the content of the message.

Write the body of the message.

Attach any relevant files if needed.

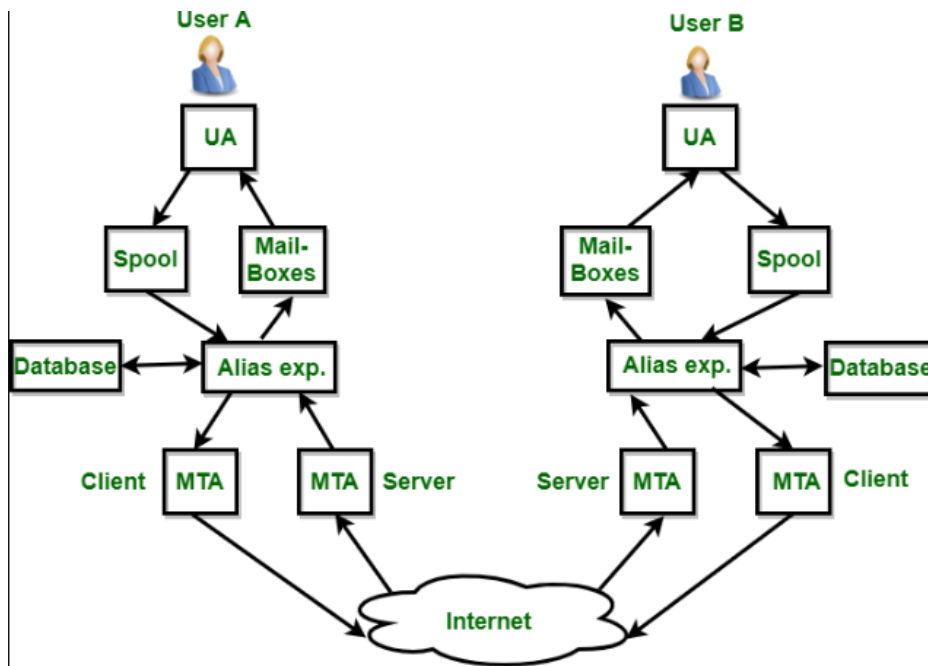
Click "Send" to deliver the message to the recipient's email server.

Emails can also include features such as cc (carbon copy) and bcc (blind carbon copy) to send copies of the message to multiple recipients, and reply, reply all, and forward options to manage the conversation.

Electronic Mail (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a message in formatted manner (mail) to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called sender and person who receives mail is called recipient. It is just like postal mail service. Components of E-Mail System : The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.

User Agent (UA) : The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.

Message Transfer Agent (MTA) : MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.



Mailbox : It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox . Access to mailbox is only to owner of mailbox.

Spool file : This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an alias, to represent several different e-mail addresses. It is known as mailing list, Whenever user have to sent a message, system checks recipient's name against alias database. If mailing list is present for defined alias, separate messages, one for each entry in the list, must be prepared and handed to MTA. If for defined alias, there is no such mailing list is present, name itself becomes naming address and a single message is delivered to mail transfer entity.

20 b) Identify and describe the DNS attacks.

Domain Name Server is a prominent building block of the Internet. It's developed as a system to convert alphabetical names into IP addresses, allowing users to access websites and exchange e-mails. DNS is organized into a tree-like infrastructure where the first level contains topmost domains, such as .com and .org. The second-level nodes contain general, traditional domain names. The 'leaf' nodes on this tree are known as hosts.

DNS works similarly to a database that is accessed by millions of computer systems in trying to identify which address is most likely to solve a user's query.

In DNS attacks, hackers will sometimes target the servers which contain the domain names. In other cases, these attackers will try to determine vulnerabilities within the system itself and exploit them for their own good.

Types of Attacks:

Denial of service (DoS): An attack where the attacker renders a computer useless (inaccessible) to the user by making a resource unavailable or by flooding the system with traffic.

Distributed denial of service (DDoS): The attacker controls an overwhelming amount of computers (hundreds or thousands) in order to spread malware and flood the victim's computer with unnecessary and overloading traffic. Eventually, unable to harness the power necessary to handle the intensive processing, the systems will overload and crash.

DNS spoofing (also known as DNS cache poisoning): An attacker will drive the traffic away from real DNS servers and redirect them to a "pirate" server, unbeknownst to the users. This may cause the corruption/theft of a user's personal data.

Fast flux: An attacker will typically spoof his IP address while performing an attack. Fast flux is a technique to constantly change location-based data in order to hide where exactly the attack is coming from. This will mask the attacker's real location, giving him the time needed to exploit the attack. Flux can be single or double or of any other variant. A single flux changes the address of the webserver while double flux changes both the address of the web server and the names of DNS serves.

Reflected attacks: Attackers will send thousands of queries while spoofing their own IP address and using the victim's source address. When these queries are answered, they will all be redirected to the victim himself.

Reflective amplification DoS: When the size of the answer is considerably larger than the query itself, a flux is triggered, causing an amplification effect. This generally uses the same method as a reflected attack, but this attack will overwhelm the user's system's infrastructure further.